

South Dakota School of Mines & Technology
Geomatics I
Spring 2026

CEE 140 and 140L
2 Credits

Instructor Information

Heidi Sieverding

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Civil-Mechanical Building (CM) Room 120

Office Hours: Tues/Thurs from 9:30-11AM, Wed from 1-3PM, and Thurs 12:30-2PM

If my door is open, please feel free to stop in and ask questions. If I am in a meeting or on a conference call, please leave me a message or email me and we can schedule an appointment. Email is typically the easiest way to reach me. I will try to respond to emails within 24 hours or on the next business day (non-holiday, non-weekend).

Course Information**Start/End Dates**

January 12 – May 8, 2026

Meeting Times and Location

Tues 12:30-1:45PM in CM 310

Labs 2-4:50PM in CM 310 on Tues. or Thurs.

Delivery Method

This is an in-person course. D2L will be used for communication, course materials, course work, and grading.

Prerequisites

None

Topics

Spatial data visualization and communication

Course Description

Students will learn to develop point cloud models and other geomatic data, visualize the 3D geometries, and translate those to scaled engineering drawings using AutoDesk and other 3D geometric modeling tools.

Student Learning Outcomes

Students successfully completing this course will have the ability to:

- Visualize objects and ideas.
- Use and understand engineering drawing terminology.
- Sketch and interpret basic civil engineering plans.
- Calculate dimensions, areas, and volumes using Civil3D.
- Compile and edit spatial data.
- Transfer civil engineering data packages and communicate spatial data.

ABET Student Outcome 7: Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

SD BOR Goals 1.2 and 1.5: Train students how to think and provide foundational skills.

This course will cover principles of engineering graphics and an introduction to computer-aided engineering drafting and map-making. This course will introduce fundamental sustainability concepts, engineering design evaluations, and iterative analyses which support sustainable design development.

Course Materials

Required Textbook(s) and Materials Engineer and Architectural scales and triangle (available in bookstore)

Engineering drafting paper (available in bookstore)

Introduction to AutoCAD 2026 for Civil Engineering Applications, Yasmin

ISBN: 9781630576769 (available through bookstore via 1st Day Access*)

*This book can be purchased independently as a paperback through Amazon or Barnes and Noble or as an ebook through Apple or Google, but will likely be cheapest through the bookstore as an ebook. Note: if you purchase an older version of the book, please be aware that content may be different.

AutoDesk Civil3D and ReCap 2026 – You will receive an email notification with a download link. If you do not receive this notification by the first day of lab, please notify Dr. Sieverding.)

Google Earth (free download: <https://www.google.com/earth/versions/>)

Technology Equipment and Skills Needed for the Course Students will be required to have a working laptop with the Microsoft Office software suite and capable of operating Civil 3D. Students should have basic computer use skills and to be able to navigate D2L and upload/download files and use tools such as MS Office programs, email, and digital textbooks.

Course Grading

Coursework

Coursework will primarily be individual, teams or partners may be assigned for some activities or assignments. Grades will be based on assignments, drawings, quizzes, final project, in-class and in-lab activities, participation, and attendance. D2L will be used to provide assignment instructions, grading rubrics, feedback, and grades. All assignments will be due at the beginning of the next class or end of the lab period with the exception of in-class or in-lab activities.

Attendance Policy

Students are expected to attend all classes and laboratory activities in person and remain until all assigned work is completed or the period is over. Attendance may be taken at any time. If a sign-in sheet will be provided, it is your responsibility to sign in during each period. If a survey or quiz is provided as a method of taking attendance, it is your responsibility to complete it during the designated time fram.

Late/Make-up Assignment Policy

Late assignments will be accepted; however, they will be assessed a 10% per week penalty. Exceptions can be granted due to circumstances such as illness at the discretion of the instructor and TA.

Academic Integrity

South Dakota Mines is committed to academic honesty and scholarly integrity. The South Dakota Board of Regents ([BOR](#)) [Policy 2.9.2](#) provides a comprehensive definition of “Academic Dishonesty”, which include cheating and plagiarism. All Instructors at South Dakota Mines are required to report allegations of academic misconduct to the Student Conduct Officer. [BOR Policy 3.4.1](#) provides detailed information regarding key definitions, policy information, prohibited conduct, and the Student Conduct process adhered to at South Dakota Mines. Any student suspected of violating academic integrity standards will be reported in accordance with the process outlined on the South Dakota Mines [website](#).

Grading and Assessment

Grading, rubrics, and feedback will be available via D2L. The grade for the lecture and lab will be combined and one D2L site will be used for all coursework.

Category Description	Percent
Lab Assignments	50%
Quizzes	15%
Final Project	20%
Attendance (Sign-in Sheets, Check-ins) and Participation	15%

Special Note Regarding Final Exams: Per South Dakota Mines Policy ([II-6-2](#)), if you are scheduled to take three or more final/last exams on the same day during finals week, you may request that the middle exam(s) of the day be rescheduled. ***You are required to make this request of your instructor(s) at least 30 days prior to the last day of regular classes.***

Grading Scale

Letter Grade	Percent
A	90 to 100%
B	80 to 89%
C	70 to 79%
D	60 to 69%
F	0 to 59%

Academic Freedom Statement

Academic Freedom is the cornerstone upon which higher education is built. Academic freedom, as defined by [BOR Policy 1.6.1](#), is fundamental to the advancement of truth, development of critical thinking, promotion of civil discourse, and contribution to the public good. Each course includes the freedom to discuss relevant matters and present various scholarly views in the classroom, as determined by the subject-matter expertise of the instructor. Students are encouraged to develop the capacity for critical thinking and to pursue the truth, debate ideas, express and evaluate their opinions, and draw conclusions. Students are free to take reasoned exception to the views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.¹

¹Language adapted from the American Association of University Professors "Joint Statement on Rights and Freedoms of Students".

Complaint Process

While we hope that every student has a meaningful and positive experience at South Dakota Mines, should a concern arise, students are encouraged to first attempt to resolve their concern directly with the person or office directly involved. Following that attempt, should the concern remain unresolved, students are encouraged to reach out to the Dean of Students office at DeanOfStudents@sdsmt.edu or 605.394.2416. Additionally, students may access the [online form](#) to submit their complaint, appeal, or grievance.

Grade Appeal Policy

In alignment with [BOR Policy 2.9.1](#), and [SDSMT Policy 2-21](#), students who wish to appeal their final course grade shall first discuss the matter with the course instructor. If the concerns are unresolved following that discussion, students may appeal to the instructor's department head or supervisor for a decision. If the student is dissatisfied with the supervisor decision, the student may then utilize the [online form](#) to submit "Appeal – Academic" for a "Grade Dispute".

Opportunity for All - Student Success Services and Support

Students are provided a one-stop source for information regarding all the services and supports to ensure success. Visit the [Opportunity Center](#) page to learn more.

Accessibility Accommodations

If you have a disability for which you are or may be requesting an accommodation, you are encouraged to contact both your instructor and the Accessibility Services office at 605.394.2533 or DisabilityServices@sdsmt.edu as early as possible in the semester.

South Dakota Board of Regents Required Syllabus Statements

The following statements may be found online in South Dakota Board of Regents Academic Affairs Council Guideline [2.7.3.A\(1\)](#):

- Freedom in Learning
- Americans with Disabilities Act
- Academic Dishonesty and Misconduct
- Acceptable Use of Technology
- Emergency Alert Communication